

Scalability & Future Plan

Date	3 July,2026
Team ID	-
Project Name	RISING WATERS:FLOOD PREDICTION SYSTEM USING ML
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	The model works only on structured data	Cannot work on satellite images which gives precise prediction by analysis of images	High
2	The model cannot handle multiple users at a time.	Multiple people having flood emergency cannot use this application	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	low	Using a Flask API to handle user load and deployment in heroku
2	Data Storage	low	Add a database to store user inputs
3	Performance	high	Use Advanced FLASK APIfor speed
4	Security	low	Use FLASK API

Future Roadmap:

Phase	Planned Feature / Enhancement
1. DATASET CHANGE	USING SATELLITE IMAGES DATSET OF FLOOD RISK AREA FOR BETTER MODEL PERFORMANCE BY ANALYSIS OF CLIMATIC CHANGES
2 .MODEL CHANGE	CHANGING THE MODEL TO CNN TO ANALYZE THE SATELLITE IMAGE FOR PRECISE PREDICTION.
3 . DEPLOYMENT FRAMEWORK CHANGE	CHANGE THE FRAMEWORK TO FASTAPI FOR HANDLING MULTIPLE USERS AND FOR SPEED
4. PLATFORM FOR REAL WORLD DEPLOYMENT	USING PLATFORMS LIKE HEROKU,PYTHON ANYWHERE TO DEPLOY THE APPLICATION FOR REAL WORD PURPOSE